

Formal Methods in Software Engineering (SE4033)

Course Instructor(s):

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Section(s): A, B, C, D, E, F, G

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 50
Total Questions: 2

Date: Apr 8, 2025

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

**Clearly state the question number in the answer script.
Misabeled questions will not be checked. All questions
must be attempted on the answer sheet in order. Any part
done with pencil or on question paper will not be marked.**

**Q-1: MCQs – Clearly indicate the correct answers on the
provided OMR sheet without cutting or overwriting. There are a
total of 30 MCQs. For each question, choose all answers that
apply. Overwriting or cutting will result in zero marks. MCQs not
solved on the OMR sheet will also receive zero marks.**

[30 marks]

Keeping syntax and usage of FOL in mind, answer the following MCQs on the provided OMR Sheet :

- 1) In a domain with people and the relation Likes(x,y), which of the following statements asserts that "Everyone likes themselves"?
A) $\forall x \text{ Likes}(x,x)$
B) $\exists x \text{ Likes}(x,x)$
C) $\forall x,y \text{ Likes}(x,y)$
D) $\forall x \neg \text{ Likes}(x,x)$
- 2) Which of the following statements ensures that the relation Knows(x,y) is symmetric (if x knows y, then y knows x)?
A) $\forall x,y (\text{Knows}(x,y) \rightarrow \text{Knows}(y,x))$
B) $\forall x,y (\text{Knows}(x,y) \wedge \text{Knows}(y,x))$
C) $\forall x,y (\text{Knows}(x,y) \vee \text{Knows}(y,x))$
D) $\forall x,y (\text{Knows}(x,x) \rightarrow \text{Knows}(y,y))$
- 3) Which formula asserts that the relation R(x,y) is reflexive (every element is related to itself)?
A) $\exists x R(x,x)$
B) $\forall x R(x,x)$
C) $\forall x,y R(x,y)$
D) $\forall x \neg R(x,x)$
- 4) Which statement ensures that the relation Ancestor(x,y) is transitive (if x is an ancestor of y and y is an ancestor of z, then x is an ancestor of z)?
A) $\forall x,y,z (\text{Ancestor}(x,y) \vee \text{Ancestor}(y,z) \rightarrow \text{Ancestor}(x,z))$

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- B) $\forall x,y,z (\text{Ancestor}(x,y) \wedge \text{Ancestor}(y,z) \rightarrow \text{Ancestor}(x,z))$
C) $\exists x,y,z (\text{Ancestor}(x,y) \wedge \text{Ancestor}(y,z))$
D) $\forall x,y,z (\text{Ancestor}(x,y) \rightarrow \text{Ancestor}(y,z))$

- 5) In a school, each student has exactly one assigned teacher. Which formula correctly models this relationship using the relation $\text{TeacherOf}(x,y)$, where x is a teacher of y ?
- A) $\forall x,y (\text{TeacherOf}(x,y) \rightarrow \exists t \text{TeacherOf}(t,y))$
B) $\forall y \exists x (\text{TeacherOf}(x,y) \wedge \forall t (\text{TeacherOf}(t,y) \rightarrow t=x))$
C) $\forall x \exists y (\text{TeacherOf}(x,y) \vee \text{TeacherOf}(y,x))$
D) $\exists x,y (\text{TeacherOf}(x,y) \wedge \forall t \text{TeacherOf}(t,y))$

- 6) A company ensures that every department has at least one employee and that each employee belongs to exactly one department. Which of the following best represents this rule in FOL?

- A) $\forall d \exists e \text{WorksIn}(e,d) \wedge \forall e \exists d \text{WorksIn}(e,d)$
B) $\forall d \exists e \text{WorksIn}(e,d) \wedge \forall e \forall d \text{WorksIn}(e,d)$ c
C) $\exists d \forall e \text{WorksIn}(e,d) \wedge \forall e \exists d \text{WorksIn}(e,d)$
D) $\forall e \exists d \text{WorksIn}(e,d) \vee \forall d \exists e \text{WorksIn}(e,d)$

- 7) An online examination system follows this rule: "Only students who have registered and paid the exam fee can access the exam portal." Which FOL formula correctly enforces this rule?

- A) $\forall x (\text{Registered}(x) \vee \text{Paid}(x) \rightarrow \text{Access}(x))$
B) $\forall x (\text{Registered}(x) \wedge \text{Paid}(x) \rightarrow \text{Access}(x))$
C) $\forall x (\text{Access}(x) \rightarrow \text{Registered}(x) \wedge \text{Paid}(x))$ b
D) $\forall x (\text{Access}(x) \rightarrow \text{Registered}(x) \vee \text{Paid}(x))$

- 8) If an SMT solver knows $a = b$ and $f(a) = c$, what can it infer about $f(b)$?

- A) $f(b) = c$
B) $f(b) \neq c$
C) No inference possible
D) $f(b) = f(a)$

- 9) What is the difference between $\forall x \exists y P(x, y)$ and $\exists y \forall x P(x, y)$?

- A) They are logically equivalent.
B) The first allows y to depend on x ; the second requires a fixed y for all x .
C) The first is satisfiable only if the second is.
D) The second implies the first, but not vice versa.

10. Let $\text{Graph}(x)$ be a predicate which denotes that x is a graph. Let $\text{Connected}(x)$ be a predicate which denotes that x is connected. Which of the following first order logic sentences **DOES NOT** represent the statement: "Not every graph is connected"?

- A) $\neg \forall x (\text{Graph}(x) \Rightarrow \text{Connected}(x))$
B) $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$
C) $\neg \forall x (\neg \text{Graph}(x) \vee \text{Connected}(x))$
D) $\forall x (\text{Graph}(x) \Rightarrow \neg \text{Connected}(x))$

11. Which of the following best describes how an SMT solver extends SAT solving techniques?

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- A) It applies brute-force search on first-order logic formulas.
- B) It converts first-order logic formulas into propositional logic and solves them directly.
- C) It integrates SAT solvers with specialized theory solvers to handle constraints efficiently.
- D) It uses machine learning models to predict satisfiable formulas.

12. Which of the following is **NOT** a valid first-order logic formula?

- A) $\forall x (P(x) \rightarrow \exists y Q(y))$
- B) $\exists x P(x) \wedge \forall y Q(y)$
- C) $P(x) \wedge Q(x) \rightarrow R$
- D) $\forall x \exists y (P(x, y) \vee Q(x) \wedge \forall z R(z))$

13. In First-Order Logic, which of the following is an example of a function symbol?

- A) $\forall x (\text{Student}(x) \rightarrow \exists y \text{Book}(y))$
- B) $f(x) = x + 1$
- C) $P(x) \rightarrow Q(x)$
- D) $\exists x \forall y \text{Buys}(x, y)$

14. Which of the following FOL sentences is NOT satisfiable?

- A) $\forall x (P(x) \rightarrow \neg P(x))$
- B) $\exists x (P(x) \wedge \neg P(x))$
- C) $\forall x \exists y (P(x) \wedge Q(y))$
- D) $\forall x (P(x) \vee \neg P(x))$

15. The formula $\forall x \exists y (P(x, y) \rightarrow P(y, x))$ expresses what kind of relationship between x and y?

- A) Reflexivity
- B) Symmetry
- C) Transitivity
- D) Anti-symmetry

16. Which of the following FOL formulas correctly expresses: "Every cat has an owner who feeds it"?

- A) $\forall x (\text{Cat}(x) \rightarrow \exists y (\text{Owner}(y) \wedge \text{Feeds}(y, x)))$
- B) $\forall x (\text{Owner}(x) \rightarrow \exists y (\text{Cat}(y) \wedge \text{Feeds}(x, y)))$
- C) $\exists x \forall y (\text{Cat}(x) \rightarrow \text{Feeds}(y, x))$
- D) $\forall x \forall y (\text{Cat}(x) \wedge \text{Feeds}(y, x) \rightarrow \text{Owner}(y))$

17. Identify the free variable(s) in the formula: $\forall x \forall y (x+z=y)$

- A) x and y
- B) y and z
- C) Only z
- D) No free variables

18. Consider the formula: $\forall x (x=y) \vee \exists z (z > x)$ Which of the following statements is correct?

- A) x and y are free variables
- B) y is a free variable, but x and z are bound

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- C) z is a free variable, but x and y are bound
- D) No free variables exist

19. According to the Theory of Equality T_E , which statement correctly represents the equivalence of predicates?

- A) $P(x)=P(y)$ for $x=y$
- B) $f(x,y)=g(x,y)$
- C) $f(x,y)=f(y,x)$
- D) $x=f(x,x)$

20. Which of the following statements is **NOT** necessarily true in the Theory of Equality T_E ?

- A) $f(x,y)=f(y,x)$
- B) $x=x$
- C) $P(x) \leftrightarrow P(y)$ for $x=y$
- D) $f(x_1, \dots, x_n)=f(y_1, \dots, y_n)$ if $x_i=y_i$ for all i

[CLO 7: Formalize specifications using Z3 with SMT-LIB syntax]

21. An embedded system enforces the following safety rule: "If the temperature is above 75°C or the pressure is above 120 PSI, and the system is in operation mode, then the system must trigger a shutdown." Which SMT-LIB assertion correctly models this safety rule?

Common code:

```
(declare-const temperature Int)
(declare-const pressure Int)
(declare-const operational Bool)
(declare-const shutdown Bool)
```

- A) (assert (=> (and (or (> temperature 75) (> pressure 120)) operational) shutdown))
- B) (assert (=> (or (> temperature 75) (> pressure 120) operational) shutdown))
- C) (assert (=> (and (or (> temperature 75) (> pressure 120) operational)) shutdown))
- D) (assert (=> (or (and (> temperature 75) operational) (and (> pressure 120) operational)) shutdown))

22. In an analytics engine, the formula used to normalize a performance score is:

$$score = \frac{((x + y)^2 - z)}{2w + 3}$$

Common code:

```
(declare-const x Int)
(declare-const y Int)
(declare-const z Int)
(declare-const w Int)
(declare-const score Int)
```

Which of the following assert statements correctly encodes the formula?

- A) (assert (= score (/ (- (* (+ x y) (+ x y)) z) (+ (* 2 w) 3))))

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- B) (assert (= score (/ (- (+ x y z)) (* 2 w 3))))
- C) (assert (= score (/ (- (* x y) z) (+ 2 (* w 3)))))
- D) (assert (= score (/ (* (- (+ x y) z) 2) (+ w 3))))

23. In a university advising system, the following policy is enforced: "Every student must be assigned to an advisor who teaches at least one course that the student is enrolled in."

Common code:

```
(declare-sort Student)
(declare-sort Advisor)
(declare-sort Course)
```

```
(declare-fun Enrolled (Student Course) Bool)
(declare-fun Teaches (Advisor Course) Bool)
(declare-fun Assigned (Student Advisor) Bool)
```

$\forall s \exists a [\text{Assigned}(s,a) \wedge \exists c [\text{Teaches}(a,c) \wedge \text{Enrolled}(s,c)]]$

Which of the following assertions correctly models this policy?

```
A) (assert (forall ((s Student))
  (exists ((a Advisor))
    (and (Assigned s a)
      (exists ((c Course))
        (and (Teaches a c) (Enrolled s c)))))))
```

```
B) (assert
  (exists ((s Student))
    (forall ((a Advisor))
      (implies (Assigned s a)
        (forall ((c Course))
          (implies (Teaches a c) (Enrolled s c)))))))
```

```
C) (assert
  (forall ((s Student))
    (forall ((a Advisor))
      (implies (Assigned s a)
        (exists ((c Course))
          (and (Enrolled s c) (Teaches a c)))))))
```

```
D) (assert
  (forall ((s Student))
    (exists ((c Course))
      (exists ((a Advisor))
        (and (Assigned s a) (Teaches a c) (Enrolled s c))))))
```

24. What does the following SMT-LIB 2 code do?

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```
(declare-const x Int)
(assert (> x 5))
(check-sat)
```

- A) Declares x as an integer and checks if $x > 5$ is satisfiable.
- B) Declares x as an integer and assigns it a value greater than 5.
- C) Declares a function x and immediately sets $x = 5$.
- D) Declares x but does nothing since there is no get-model command.

25. Which SMT-LIB code correctly asserts $(x - y = 5) \wedge (x > 0)$?

```
A) (declare-const x Int)
    (declare-const y Int)
    (assert (= (- x y) 5))
    (assert (> x 0))
    (check-sat)
```

```
B) (declare-const x Int)
    (declare-const y Int)
    (assert ((= x - y 5) (> x 0)))
```

```
C) (declare-const x Int)
    (declare-const y Int)
    (assert (= x (- y 5)))
    (assert (> x 0))
```

D) None of these

26. In Z3's SMT-LIB2 syntax, which of the following correctly declares a Boolean constant p and asserts the formula $p \wedge \neg p$?

```
A) (declare-const p Bool)
    (assert (and p (not p)))
```

```
B) (declare-fun p () Bool)
    (assert (and p (!p)))
```

```
C) (declare-const p Int)
    (assert (/\ p (not p)))
```

```
D) (declare-const p Bool)
    (assert (implies p (not p)))
```

27. Which of the following correctly asserts that "For all integers x, if x is positive, then x squared is greater than x"?

```
A) (assert (forall ((x Int)) (=> (> x 0) (> (* x x) x))))
```

```
B) (forall ((x Int)) (=> (> x 0) (> (* x x) x)))
```

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- C) (assert (forall (x Int) (implies (> x 0) (> (* x x) x))))
- D) (assert (exists ((x Int)) (=> (> x 0) (> (* x x) x))))

28. What will be the result of the following Z3's SMT-LIB2 script?

```
(declare-const x Int)
(assert (< x 0))
(assert (> x 0))
(check-sat)
```

- A) sat
- B) **unsat**
- C) unknown
- D) error

29. Which of the following statements correctly describes the difference between declare-fun and define-fun in Z3 SMT-LIB2?

- A) **declare-fun only specifies the function signature, while define-fun provides both the signature and the function body.**
- B) declare-fun allows defining function bodies, whereas define-fun only declares function names.
- C) define-fun is used for declaring constants, while declare-fun is used for defining predicates.
- D) Both declare-fun and define-fun are interchangeable and have the same functionality.

30. What is the purpose of the assert command in Z3 SMT-LIB2?

- A) It declares a new function in the SMT solver.
- B) It defines a function with a body.
- C) **It adds constraints or assumptions to the solver.**
- D) It checks whether a given formula is satisfiable.

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Formal Methods in SE

First Order Logic

A B C D

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A B C D

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A B C D

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A B C D

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Z3 SMT-LIB

A B C D

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A B C D

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[CLO 8: Apply formal methods to ensure code correctness]

Q2: Provide Static Single Assignment formulation for the following piece of code. Unroll the loop twice and use ϕ notation to denote conditions. Specify the return values corresponding to SSA assignments.

[20 marks]

You may assume for `arr[3] = arr0, arr1, arr2`.

```
int[] magic(int arr[], int n){
    int j, temp = 0;

    for (j = 0; j < n - 1; j++) {

        if (arr[j] > arr[j + 1]) {

            temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;

        }
        else {
            arr[j] = j
        }
    }

    return arr;
}
```

Best of Luck 😊